EXERCISE-I (Conceptual Questions) Build Up Your Understanding

ELECTROLYTIC CONDUCTANCE

1. Strong electrolytes are those which :
(1) dissolve readily in water
(2) conduct electricity
(3) dissociate into ions even at high concentration
(4) dissociate into ions at high dilution.

EC0001

2. Molten sodium chloride conducts electricity due to the presence of :
(1) free electrons
(2) free ions
(3) free molecules
(4) free atoms of Na and Cl

EC0002

3. Electrolytic conduction is due to the movement of :
(1) molecules (2) atoms
(3) ions (4) electrons

EC0003

4. Which of the following solutions of KCl has the lowest value of equivalent conductance ?
(1) 1 M (2) 0.1 M
(3) 0.01 M (4) 0.001 M

EC0004

5. If the specific resistance of a solution of concentration C geq L^{-1} is R, then its equivalent conductance is :

(1) $\frac{100R}{C}$ (2) $\frac{RC}{1000}$

(3) $\frac{1000}{RC}$ (4) $\frac{C}{1000R}$

EC0006

6. The specific conductances in $ohm^{-1}\ cm^{-1}$ of four electrolytes P, Q, R and S are given in brackets :
P (5.0×10^{-5}) Q (7.0×10^{-8})
R (1.0×10^{-10}) S (9.2×10^{-3})
The one that offers highest resistance to the passage of electric current is
(1) P (2) S (3) R (4) Q

EC0007

7. The specific conductance of a salt of 0.01 M concentration is 1.061×10^{-4} S cm^{-1}. Molar conductance of the same solution will be :
(1) 1.061×10^{-4} S $cm^2\ mol^{-1}$
(2) 1.061 S $cm^2\ mol^{-1}$
(3) 10.61 S $cm^2\ mol^{-1}$
(4) 106.1 S $cm^2\ mol^{-1}$

EC0008

8. Which statement is not correct :–
(1) Conductance of an electrolytic solution increases with dilution
(2) Conductance of an electrolytic solution decreases with dilution
(3) Specific conductance of an electrolytic solution decreases with dilution
(4) Equivalent conductance of an electrolytic solution increases with dilution.

EC0011

9. The resistance of 0.01 N solution of an electrolyte was found to be 210 ohm at 298 K using a conductivity cell of cell constant 0.66 cm^{-1}. The equivalent conductance of solution is :–
(1) 314.28 mho $cm^2\ eq^{-1}$
(2) 3.14 mho $cm^2\ eq^{-1}$
(3) 314.28 $mho^{-1}\ cm^2\ eq^{-1}$
(4) 3.14 $mho^{-1}\ cm^2\ eq^{-1}$

EC0012

10. Electrolytic conduction differs from metallic conduction from the fact that in the former
(1) The resistance increases with increasing temperature
(2) The resistance decreases with increasing temperature
(3) The resistance remains constant with increasing temperature
(4) The resistance is independent of the length of the conductor

EC0013

11. The specific conductance of a 0.01 M solution of KCl is 0.0014 $ohm^{-1}\ cm^{-1}$at 25° C. Its equivalent conductance ($cm^2\ ohm^{-1}\ eq^{-1}$) is :–
(1) 140 (2) 14
(3) 1.4 (4) 0.14

EC0014

12. Which one of the following is wrong :–
(1) Specific conductance increases on dilution.
(2) Specific conductance decreases on dilution.
(3) Equivalent conductance increases on dilution.
(4) Molar conductance increases on dilution.

EC0015

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KOHLRAUSCH LAW

13. At infinite dilution, the equivalent conductances of CH_3COONa, HCl and CH_3COOH are 91, 426 and 391 mho cm^2 eq^{-1} respectively at 25 °C. The eq. conductance of NaCl at infinite dilution will be :

(1) 126 (2) 209 (3) 391 (4) 908

EC0016

14. For HCl solution at 25 °C, equivalent conductance at infinite dilution is 425 ohm^{-1} cm^2 eq^{-1}. The specific conductance of a solution of HCl is 3.825 ohm^{-1} cm^{-1}. If the degree of dissociation is 90%, the normality of the solution is :-

(1) 0.90 N (2) 1.0 N

(3) 10 N (4) 1.2 N

EC0019

15. The molar conductivities $\wedge^0_{NaOAc}$ and $\wedge^0_{HCl}$ at infinite dilution in water at 25°C are 91.0 and 426.2 S cm^2 mol^{-1} respectively. To calculate $\wedge^0_{HOAc}$ the additional value required is :

(1) $\wedge^0_{NaCl}$ (2) $\wedge^0_{H_2O}$ (3) $\wedge^0_{KCl}$ (4) $\wedge^0_{NaOH}$

EC0020

16. The molar conductance of $AgNO_3$, AgCl and NaCl at infinite dilution are 116.5, 121.6 and 110.3 S cm^2 mol^{-1} respectively. The molar conductance of $NaNO_3$ is :

(1) 111.4 S cm^2 mol^{-1} (2) 105.2 S cm^2 mol^{-1}

(3) 130.6 S cm^2 mol^{-1} (4) 150.2 S cm^2 mol^{-1}

EC0021

17. The conductivity of a saturated solution of $BaSO_4$ is 3.06×10^{-6} ohm^{-1} cm^{-1} and its molar conductance is 1.53 ohm^{-1} cm^2 mol^{-1}. The K_{sp} of $BaSO_4$ will be

(1) 4×10^{-12} (2) 2.5×10^{-9}

(3) 2.5×10^{-13} (4) 4×10^{-6}

EC0022

18. Kohlrausch's law states that at :-

(1) Infinite dilution, each ion makes definite contribution to conductance of an electrolyte whatever be the nature of the other ion of the electrolyte.

(2) Infinite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte whatever be the nature of the other ion of the electrolyte.

(3) Finite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte whatever be the nature of the other ion of the electrolyte.

(4) Infinite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte depending on the nature of the other ion of electrolyte.

EC0099

GALVANIC CELL

19. In the galvanic cell

Cu(s) | Cu^{2+} (1 M) || Ag^+ (1 M) | Ag(s)

the electrons will travel in the external circuit :

(1) from Ag to Cu

(2) from Cu to Ag

(3) electrons do not travel in the external circuit

(4) in any direction

EC0023

20. The direction of current in the Daniell cell when Zn and Cu electrodes are connected is :

(1) from Cu to Zn in the cell

(2) from Cu to Zn out side the cell

(3) from Zn to Cu outside the cell

(4) in any direction in the cell

EC0024

21. The equation representing the process by which standard reduction potential of zinc can be defined is

(1) $Zn^{2+}(s) + 2e^- \longrightarrow Zn(s)$

(2) $Zn(g) \longrightarrow Zn^{2+}(g) + 2e^-$

(3) $Zn^{2+}(g) + 2e^- \longrightarrow Zn(s)$

(4) $Zn^{2+}(aq.) + 2e^- \longrightarrow Zn(s)$

EC0025

22. A standard hydrogen electrode has zero electrode potential because :

(1) Hydrogen is easiest to oxidize.

(2) This electrode potential is assumed to be zero.

(3) Hydrogen atom has only one electron.

(4) Hydrogen is the lightest element.

EC0026

23. Which is not true for a standard hydrogen electrode ?

(1) The hydrogen ion concentration is 1 M.

(2) Temperature is 25°C.

(3) Pressure of hydrogen is 1 bar.

(4) It contains a metallic conductor which does not adsorb hydrogen.

EC0027

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24. E° for the half cell
Zn^{2+} | Zn is –0.76 V. E.M.F. of the cell
Zn | Zn^{2+} (1M) || $2H^+$ (1M) | H_2 (1 atm) is :
(1) –0.76 V (2) +0.76 V
(3) –0.38 V (4) + 0.38 V

EC0028

25. Cu(s) | Cu^{+2}(1 M) || Zn^{+2}(1 M) | Zn(s)
A cell represented above should have emf.
(1) Positive
(2) Negative
(3) Zero
(4) Cannot be predicted

EC0030

26. Given electrode potentials :
$Fe^{3+} + e \longrightarrow Fe^{2+}$; E° = 0.771 V
$I_2 + 2e \longrightarrow 2I^-$; E° = 0.536 V
E°_{cell} for the cell reaction
$2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$ is -
(1) (2 ×0.771 – 0.536) = 1.006 V
(2) (0.771 – 0.5 ×0.536) = 0.503 V
(3) 0.771 – 0.536 = 0.235 V
(4) 0.536 – 0.771 = –0.235 V

EC0031

27. Which of the following is not an anodic reaction :–
(1) $Ag^+ \rightarrow Ag - e^-$
(2) $Cu \rightarrow Cu^{2+} + 2e^-$
(3) $Fe^{2+} \rightarrow Fe^{3+} + e^-$
(4) $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$

EC0032

28. Which of the following statements is correct :–
(1) Oxidation occurs at anode in both galvanic and electrolytic cell.
(2) Reduction occurs at anode in both galvanic and electrolytic cell
(3) Reduction occurs at anode in electrolytic cell where as oxidation occurs at cathode in galvanic cell
(4) Oxidation occurs at anode in electrolytic cell where as reduction occurs at anode in a galvanic cell,

EC0033

29. Other things being equal, the life of a Daniell cell may be increased by :–
(1) Keeping low temperature
(2) Using large copper electrode
(3) Decreasing concentration of copper ions
(4) Using large zinc electrodes

EC0034

ELECTROCHEMICAL SERIES

30. Zn can not displace following ions from their aqueous solution :-
(1) Ag^+ (2) Cu^{2+}
(3) Fe^{2+} (4) Na^+

EC0035

31. The standard reduction potentials at 25 ℃ for the following half reactions are given against each :-
Zn^{2+} (aq) + $2e^-$ $\rightleftharpoons$ Zn(s), –0.762 V
Cr^{3+} (aq) + $3e^-$ $\rightleftharpoons$ Cr(s), –0.740 V
$2H^+ + 2e^-$ $\rightleftharpoons$ H_2(g), 0.00 V
$Fe^{3+} + 2e^-$ $\rightleftharpoons$ Fe^{2+}, 0.77 V
Which is the strongest reducing agent ?
(1) Zn (2) Cr
(3) H_2(g) (4) Fe^{2+} (aq)

EC0036

32. Using the standard electrode potential values given below, decide which of the statements, I, II, III and IV are correct. Choose the right answer from (1), (2), (3) and (4).
$Fe^{2+} + 2e^-$ $\rightleftharpoons$ Fe ; E° = –0.44 V
$Cu^{2+} + 2e^-$ $\rightleftharpoons$ Cu ; E° = +0.34 V
$Ag^+ + e^-$ $\rightleftharpoons$ Ag ; E° = +0.80 V
I. Copper can displace iron from $FeSO_4$ solution.
II. Iron can displace copper from $CuSO_4$ solution.
III. Silver can displace copper from $CuSO_4$ solution.
IV. Iron can displace silver from $AgNO_3$ solution.
(1) I and II (2) II and III
(3) II and IV (4) I and IV

EC0041

33. The standard electrode potentials for the elements A, B and C are 0.68, –2.50 and 0.50 V respectively. The order of their reducing power is :
(1) A > B > C (2) A > C > B
(3) C > B > A (4) B > C > A

EC0042

34. The oxidation potential of Zn, Cu, Ag, H_2 and Ni electrodes are 0.76 V, –0.34 V, –0.80 V, 0 V, 0.55 V respectively. Which of the following reaction will provide maximum voltage ?
(1) $Zn + Cu^{2+} \longrightarrow Cu + Zn^{2+}$
(2) $Zn + 2Ag^+ \longrightarrow 2Ag + Zn^{2+}$
(3) $H_2 + Cu^{2+} \longrightarrow 2H^+ + Cu$
(4) $H_2 + Ni^{2+} \longrightarrow 2H^+ + Ni$

EC0043

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35. The standard reduction potential at 25 °C of Li^+ / Li, Ba^{2+} / Ba, Na^+ / Na and Mg^{2+} / Mg are –3.05 V, –2.73 V, –2.71 V and –2.37 V respectively. Which one of the following is the strongest oxidising agent ?
(1) Na^+ (2) Li^+ (3) Ba^{2+} (4) Mg^{2+}

EC0045

36. A gas X at 1 atm is bubbled through a solution containing a mixture of 1 M Y^- and 1 M Z^- at 25 °C. If the reduction potential of Z > Y > X then :
(1) Y will oxidise X and not Z
(2) Y will oxidise Z and not X
(3) Y will oxidise both X and Z
(4) Y will reduce both X and Z

EC0046

37. The standard electrode potential of Zn, Ag and Cu electrodes are –0.76 V, 0.80 V and 0.34 V respectively, then :
(1) Ag can oxidise Zn and Cu
(2) Ag can reduce Zn^{2+} and Cu^{2+}
(3) Zn can reduce Ag^+ and Cu^{2+}
(4) Cu can oxidise Zn and Ag

EC0047

38. Standard reduction potentials of four metal electrodes are
A = – 0.250 V , B = – 0.140 V
C = – 0.126 V , D = – 0.402 V
The metal that displaces A from aqueous solution of its compounds is :-
(1) B (2) C
(3) D (4) None of the above

EC0049

39. The following four colourless salt solutions are placed in separate test tubes and a strip of Cu is placed in each solution. Which solution finally turns blue :–
(1) $Zn(NO_3)_2$ (2) $Mg(NO_3)_2$
(3) KNO_3 (4) $AgNO_3$

EC0050

40. Which of the following displacement does not occur
(1) $Zn + 2H^+ \rightarrow Zn^{2+} + H_2 \uparrow$
(2) $Fe + 2Ag^+ \rightarrow Fe^{2+} + Ag \downarrow$
(3) $Cu + Fe^{2+} \rightarrow Cu^{2+} + Fe \downarrow$
(4) $Zn + Pb^{2+} \rightarrow Zn^{2+} + Pb \downarrow$

EC0051

41. On the basis of the following E° values, the strongest oxidizing agent is :-
$[Fe(CN)_6]^{4-} \rightarrow [Fe(CN)_6]^{3-} + e^{-1}, E° = -0.35$ V
$Fe^{2+} \rightarrow Fe^{3+} + e^{-1}$; E° = – 0.77 V
(1) Fe^{3+} (2) $[Fe(CN)_6]^{3-}$
(3) $[Fe(CN)_6]^{4-}$ (4) Fe^{2+}

EC0098

NERNST EQUATION

42. $E°(Ni^{2+}/Ni) = -0.25$ V
$E°(Au^{3+} / Au) = 1.50$ V
The emf of the voltaic cell.
Ni | Ni^{2+} (1.0 M) || Au^{3+} (1.0 M) | Au is :
(1) 1.25 V (2) –1.75 V
(3) 1.75 V (4) 4.0 V

EC0052

43. The emf of the cell
Tl(s) | Tl^+ (0.0001 M) || Cu^{2+} (0.01M) | Cu(s) is 0.83 V
The emf of this cell will be increased by :–
(1) Increasing the concentration of Cu^{+2} ions
(2) Decreasing the concentration of Tl^+
(3) Increasing the concentration of both
(4) (1) & (2) both

EC0054

44. Which of the following represents the electrode potential of silver electrode dipped into 0.1 M $AgNO_3$ solution at 25° C ?
(1) $E°_{red}$ (2) $(E°_{red} + 0.059)$
(3) $(E°_{ox} - 0.059)$ (4) $(E°_{red} - 0.059)$

EC0056

45. The electrode potential of a hydrogen electrode dipped in solution of pH = 1 is
(1) 0.059 V (2) 0.00 V
(3) –0.059 V (4) 0.59 V

EC0057

46. Consider the reaction
$Cl_2(g) + 2Br^-(aq) \longrightarrow 2Cl^-(aq) + Br_2(\ell)$
The emf of the cell when
$[Cl^-] = [Br^-] = 0.01$ M and Cl_2 gas at 0.25 atm pressure will be (E° for the above reaction is = 0.29 V)
(1) 0.54 V (2) 0.272 V
(3) 0.29 V (4) –0.29 V

EC0058

47. The standard emf for the cell reaction
$Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$ is 1.10 V at 25 °C. The emf for the cell reaction when 0.1 M Cu^{2+} and 0.1 M Zn^{2+} solution are used at 25 °C is :
(1) 1.10 V (2) 0.110 V
(3) –1.10 V (4) –0.110 V

EC0059

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48. E° for $F_2 + 2e^- \rightarrow 2F^-$ is 2.8 V,
E° for $\frac{1}{2}F_2 + e^- \rightarrow F^-$ is ?
(1) 2.8 V (2) 1.4 V (3) –2.8 V (4) –1.4 V

EC0060

49. How much will the potential of $Zn \mid Zn^{2+}$ change if the solution of Zn^{2+} is diluted 10 times
(1) increases by 0.03 V
(2) decreases by 0.03 V
(3) increases by 0.059 V
(4) decreases by 0.059 V

EC0062

50. How much will the potential of a hydrogen electrode change when its solution initially at pH = 0 is neutralised to pH = 7 ?
(1) increases by 0.059 V
(2) decreases by 0.059 V
(3) increases by 0.41 V
(4) decreases by 0.41 V

EC0063

51. Which of the following will increase the voltage of the cell with following cell reaction
$Sn_{(s)} + 2Ag^+_{(aq)} \rightarrow Sn^{+2}_{(aq)} + 2Ag_{(s)}$
(1) Increase in the size of silver rod
(2) Increase in the concentration of Sn^{+2} ions
(3) Increase in the concentration of Ag^+ ions
(4) Decrease in the concentration of Ag^+ ions

EC0064

52. E° for the reaction $Fe + Zn^{2+} \rightarrow Zn + Fe^{2+}$ is –0.35 V. The given cell reaction is :
(1) spontaneous (2) non-spontaneous
(3) in equilibrium (4) can't say anything

EC0065

53. For a reaction - $A(s) + 2B^+ \rightleftharpoons A^{2+} + 2B(s)$
K_C has been found to be 10^{12}. The E°_{cell} is:
(1) 0.354 V (2) 0.708 V
(3) 0.0098 V (4) 1.36 V

EC0066

54. The standard electrode potential (E°) for OCl^-/Cl^- and $Cl^- / \frac{1}{2} Cl_2$ respectively are 0.94 V and –1.36 V. The E° value of $OCl^- / \frac{1}{2} Cl_2$ will be :
(1) –2.20 V (2) –0.42 V
(3) 0.52 V (4) 1.04 V

EC0067

55. The hydrogen electrode is dipped in a solution of pH = 3 at 25 °C. The electrode potential of the half cell would be :
(1) 0.177 V (2) – 0.177 V
(3) 0.087 V (4) 0.059 V

EC0069

56. What is the potential of the cell containing two hydrogen electrodes as represented below
$Pt; H_2(g) \mid H^+(10^{-8})M \parallel H^+(0.001\ M) \mid H_2(g).Pt$;
(1) – 0.295 V (2) – 0.0591 V
(3) 0.295 V (4) 0.0591 V

EC0070

57. Consider the cell $Cu \mid Cu^{+2} \parallel Ag^+ \mid Ag$. If the concentration of Cu^{+2} and Ag^+ ions becomes ten times, then the emf of the cell will :–
(1) Becomes 10 times
(2) Remains same
(3) Increases by 0.0295 V
(4) Decreases by 0.0295 V

EC0071

58. The emf of the cell
$Ni \mid Ni^{+2}(1.0\ M) \parallel Au^{+3}(0.1M) \mid Au$
[E° for $Ni^{+2} \mid Ni = –0.25$ V, E° for $Au^{+3} \mid Au = 1.50$ V] is given as:-
(1) 1.25 V (2) – 1.75 V
(3) 1.78 V (4) 1.73 V

EC0072

59. The equilibrium constant (in approx) of the cell reaction :
$Cu(s) + 2Ag^+(aq.) \rightleftharpoons Cu^{+2}(aq.) + 2Ag(s)$
if $E^\circ_{cell} = 0.465$ V at 298 K is :-
(1) 2.0×10^{10} (2) 3.16×10^{12}
(3) 3.16×10^{15} (4) 4×10^{10}

EC0097

60. The emf of the cell in which the following reaction
$Zn(s) + Ni^{2+}\ (a = 0.1) \rightleftharpoons Zn^{2+}\ (a = 1.0) + Ni(s)$
occurs, is found to be 0.5105 V at 298 K. The standard e.m.f. of the cell is :-
(1) –0.5105 V (2) 0.5400 V
(3) 0.4810 V (4) 0.5696 V

EC0053

61. If $E^\circ_{Fe^{+2}/Fe} = -0.441$V and $E^\circ_{Fe^{+3}/Fe^{+2}} = 0.771$V the standard EMF of the reaction $Fe + 2Fe^{+3} \rightarrow 3Fe^{+2}$ will be :
(1) 0.330 V (2) 1.653 V
(3) 1.212 V (4) 0.111 V

EC0093

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ELECTROLYSIS

62. When an electric current is passed through acidified water, 112 mL of hydrogen gas at STP collects at the cathode in 965 s. The current passed in ampere is :
(1) 1.0 (2) 0.5
(3) 0.1 (4) 2.0

EC0073

63. Two electrolytic cells one containing acidified ferrous chloride and another acidified ferric chloride are connected in series. The ratio of iron deposited at cathodes in the two cells when electricity is passed through the cells will be :
(1) 3 :1 (2) 2 : 1 (3) 1 : 1 (4) 3 : 2

EC0074

64. A current of 9.65 A flowing for 10 minute deposits 3.0 g of a metal. The equivalent weight of the metal is :
(1) 10 (2) 30 (3) 50 (4) 96.5

EC0075

65. How many coulombs of electric charge are required for the oxidation of 1 mol of H_2O to O_2 ?
(1) 9.65×10^4 C (2) 4.825×10^5 C
(3) 1.93×10^5 C (4) 1.93×10^4 C

EC0077

66. On passing 10800 C through electrolytic solution, 2.977 g of metal (atomic mass 106.4 g mol^{-1}) was deposited, the charge on the metal cation is -
(1) +4 (2) +3 (3) +2 (4) +1

EC0079

67. On passing electricity through dilute H_2SO_4 solution the amount of substance liberated at the cathode and anode are in the ratio :
(1) 1 : 8 (2) 8 : 1
(3) 16 : 1 (4) 1 : 16

EC0080

68. During electrolysis of fused calcium hydride, the hydrogen is produced at :
(1) Cathode
(2) Anode
(3) Hydrogen is not liberated at all
(4) H_2 produced reacts with oxygen to form water

EC0081

69. A silver cup is plated with silver by passing 965 A current for one second, the mass of Ag deposited is :–(At. wt. of Ag = 107.87)
(1) 9.89 g. (2) 107.87 g.
(3) 1.0787 g. (4) 100.2 g.

EC0082

70. When electricity is passed through a solution of $AlCl_3$, 13.5 g Al is deposited. The number of Faradays must be :–
(1) 5.0 (2) 1.0 (3) 1.5 (4) 3.0

EC0083

71. A solution of sodium sulphate in water is electrolysed using inert electrodes. The product at the cathode and anode are respectively :–
(1) H_2, SO_2 (2) O_2, H_2
(3) O_2, Na (4) H_2, O_2

EC0084

72. One Faraday of electricity will liberate one mole of the metal from the solution of
(1) Auric chloride ($AuCl_3$) (2) Silver nitrate
(3) Calcium chloride (4) Copper sulphate

EC0085

73. When 96500 C of electricity are passed through barium chloride solution, the amount of barium deposited will be :-
(1) 0.5 mol (2) 1.0 mol
(3) 1.5 mol (4) 2.0 mol

EC0086

74. A factory produces 40 kg of calcium in two hours by electrolysis. How much aluminium can be produced by the same current in two hours :–
(1) 22 kg (2) 18 kg
(3) 9 kg (4) 27 kg

EC0087

75. What would be the ratio of moles of Ag, Cu, Fe that would be deposited by passage of same quantity of electricity through solutions of salts containing Ag^+, Cu^{+2}, Fe^{+3} :-

(1) 1 : 1 : 1 (2) $1 : \frac{1}{2} : \frac{1}{3}$

(3) $\frac{1}{3} : \frac{1}{2} : 1$ (4) 1 : 2 : 3

EC0088

76. Electrolysis of aq. $CuSO_4$ causes :–
(1) An increase in pH
(2) A decrease in pH
(3) Either decrease or increase in pH
(4) None

EC0089

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77. The passage of current liberates H_2 at cathode and Cl_2 at anode. The solution is :–

(1) $CuSO_4$ (aq) (2) $CuCl_2$ (aq.)
(3) NaCl (aq.) (4) Water

EC0090

COMMERCIAL CELLS

78. When lead accumulator is charged, it acts as :

(1) an electrolytic cell (2) a galvanic cell
(3) a Daniel cell (4) none of the above

EC0091

79. The thermodynamic efficiency of cell is given by -

(1) $\frac{\Delta H}{\Delta G} \times 100$ (2) $\frac{nFE}{\Delta G} \times 100$

(3) $-\frac{nFE}{\Delta H} \times 100$ (4) Zero

EC0169

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EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**3**	**2**	**3**	**1**	**3**	**3**	**3**	**2**	**1**	**2**	**1**	**1**	**1**	**3**	**1**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**2**	**4**	**2**	**2**	**2**	**4**	**2**	**4**	**2**	**2**	**3**	**1**	**1**	**4**	**4**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**1**	**3**	**4**	**2**	**4**	**1**	**3**	**3**	**4**	**3**	**1**	**3**	**4**	**4**	**3**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**2**	**1**	**1**	**1**	**4**	**3**	**2**	**1**	**3**	**2**	**3**	**3**	**4**	**3**	**2**
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	**3**	**1**	**4**	**3**	**3**	**1**	**1**	**2**	**3**	**3**	**4**	**2**	**1**	**2**	**2**
Que.	76	77	78	79											
Ans.	**2**	**3**	**1**	**3**											

EXERCISE-III (Analytical Questions)

Master Your Understanding

1. Specific conductance of 0.1 M Nitric acid is 6.3×10^{-2} ohm^{-1} cm^{-1}. The molar conductance of the solution is :

(1) 630 ohm^{-1} cm^2 mol^{-1}

(2) 315 ohm^{-1} cm^2 mol^{-1}

(3) 100 ohm^{-1} cm^2 mol^{-1}

(4) 6.300 ohm^{-1} cm^2 mol^{-1}

(5) 63.0 ohm^{-1} cm^2 mol^{-1}

EC0140

2. The highest electrical conducitivity of the following aqueous solution is of

(1) 0.1 M fluoroacetic acid

(2) 0.1 M difluoroacetic acid

(3) 0.1 M acetic acid

(4) 0.1 M chloroacetic acid

EC0141

3. Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1 M is 100 Ω. The conductivity of this solution is 1.29 S m^{-1}. Resistance of the same cell when filled with 0.02 M of the same solution is 520 Ω. The molar conductivity of 0.02 M solution of the electrolyte will be.

(1) 124×10^{-4} S m^2 mol^{-1}

(2) 1240×10^{-4} S m^2 mol^{-1}

(3) 1.24×10^{-4} S m^2 mol^{-1}

(4) 12.4×10^{-4} S m^2 mol^{-1}

EC0142

4. Resistance of 0.2 M solution of an electrolyte is 50 Ω. The specific conductance of the solution is 1.3 S m^{-1}. If resistance of the 0.4 M solution of the same electrolyte is 260 Ω then its molar conductivity is:-

(1) 6250 S m^2 mol^{-1}

(2) 6.25×10^{-4} S m^2 mol^{-1}

(3) 625×10^{-4} S m^2 mol^{-1}

(4) 62.5 S m^2 mol^{-1}

EC0144

5. The equivalent conductance of NaCl at concentration C and at infinite dilution are λ_C and λ^∞, respectively. The correct relationship between λ_C and λ^∞ is given by :

(where the constant B is positive)

(1) $\lambda_C = \lambda^\infty - (B)\sqrt{C}$ (2) $\lambda_C = \lambda^\infty + (B)\sqrt{C}$

(3) $\lambda_C = \lambda^\infty + (B)\, C$ (4) $\lambda_C = \lambda^\infty - (B)\, C$

EC0145

6. Which of the following statements is TRUE for the electrochemical Daniel cell :

(1) Electrons flow from copper electrode to zinc electrode.

(2) Current flows from zinc electrode to copper electrode.

(3) Cations move toward copper electrode.

(4) Cations move toward zinc electrode.

EC0147

7. Standard reduction electrode potential of three metal electrodes A, B and C are respectively + 0.5 V, – 3.0 V and –1.2 V. The reducing power of these metals are :

(1) C > B > A (2) A > C > B

(3) B > C > A (4) A > B > C

EC0149

8. Consider the following E^0 values

$E^0_{Fe^{3+}/Fe^{2+}} = +0.77$ V

$E^0_{Sn^{2+}/Sn} = -0.14$ V

Under standard conditions the potential for the reaction

$Sn(s) + 2Fe^{3+}(aq) \longrightarrow 2Fe^{2+}(aq) + Sn^{2+}(aq)$ is

(1) 0.91 V (2) 1.40 V

(3) 1.68 V (4) 0.63 V

EC0150

9. For the redox reaction :

$Zn(s) + Cu^{2+}$ (0.1M) $\rightarrow Zn^{2+}$ (1M) + Cu(s) taking place in a cell,

E°_{Cell} is 1.10 V. E_{Cell} for the cell will be

$\left(2.303\frac{RT}{F} = 0.0591\right)$

(1) 1.07 V (2) 0.82 V

(3) 2.14 V (4) 1.80 V

EC0151

10. In a cell that utilises the reaction

$Zn(s) + 2H^+(aq.) \rightleftharpoons Zn^{2+}(aq) + H_2(g)$ addition of H_2SO_4 to cathode compartment, will

(1) increase the E_{cell} and shift equilibrium to the right

(2) lower the E_{cell} and shift equilibrium to the right

(3) lower the E_{cell} and shift equilibrium to the left

(4) increase the E_{cell} and shift equilibrium to the left

EC0152

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11. The e.m.f. of a Daniell cell at 298 K is E_1.

$Zn \mid ZnSO_4 \mid\mid CuSO_4 \mid Cu$

(0.01M) (1.0M)

When the concentration of $ZnSO_4$ is 1.0 M and that of $CuSO_4$ is 0.01 M, the e.m.f. changed to E_2. What is the relationship between E_1 and E_2

(1) $E_1 > E_2$ (2) $E_1 < E_2$

(3) $E_1 = E_2$ (4) $E_2 = 0 \neq E_1$

EC0153

12. The reduction potential of hydrogen half-cell will be negative if :-

(1) $p(H_2)$ = 2 atm and $[H^+]$ = 1.0 M

(2) $p(H_2)$ = 2 atm and $[H^+]$ = 2.0 M

(3) $p(H_2)$ = 1 atm and $[H^+]$ = 2.0 M

(4) $p(H_2)$ = 1 atm and $[H^+]$ = 1.0 M

EC0154

13. For a spontaneous reaction; ΔG, equilibrium constant (K) and E°_{Cell} will be respectively

(1) –ve, < 1, –ve (2) –ve, >1, –ve

(3) –ve, >1, +ve (4) +ve, >1, –ve

EC0156

14. The metal that cannot be obtained by electrolysis of an aqueous solution of its salts is:

(1) Cu (2) Cr (3) Ag (4) Ca

EC0160

15. During electrolysis of a solution of $AgNO_3$, 9650 C of charge is passed through the electroplating bath. The mass of silver deposited on the cathode will be :

(1) 21.6 g (2) 108 g (3) 1.08 g (4) 10.8 g

EC0161

16. Aluminium oxide may be electrolysed at 1000 °C to furnish aluminium metal . The cathode reaction is

$Al^{3+} + 3e^- \longrightarrow Al$

The amount of charge required to prepare 5.12 kg of aluminium metal by this method would be :-

(1) 5.49×10^4 C

(2) 5.49×10^1 C

(3) 5.49×10^7 C

(4) 1.83×10^7 C

EC0162

17. A concentration cell involving a metal M is :

$M_{(s)} \mid M^+_{(aq)}(0.05M) \mid\mid M^+_{(aq)}(1M) \mid M_{(s)}$.

For this cell ;

(1) $E_{cell} > 0$; $\Delta G < 0$

(2) $E_{cell} = 0$; $\Delta G = 0$

(3) $E_{cell} > 0$; $\Delta G > 0$

(4) $E^\circ_{cell} > 0$; $\Delta G^\circ < 0$

EC0196

18. $M^{+n}(aq) + ne^- \longrightarrow M(s)$; $E^\circ_{M^{+n}/M} = xV$

$N^{+n}(aq) + ne^- \longrightarrow N(s)$; $E^\circ_{N^{+n}/N} = yV$

if x > y then :

(I) $M^{+n}_{(aq)}$ is better oxidising agent compared to $N^{+n}_{(aq)}$

(II) N(s) is better reducing agent compared to M(s)

(III) $N^{+n}_{(aq)}$ is better oxidising agent compared to $M^{+n}_{(aq)}$

(IV) M(s) is better reducing agent compared to N(s)

(1) I and III (2) II, III

(3) I and II (4) III, IV

EC0197

19. Which of the following is incorrect for

$Cu(s) \mid Cu^{+2}(aq, 1M) \mid\mid Ag^+(aq, 1M) \mid Ag(s)$

(1) $E^\circ_{cell} = E^\circ_{Cu/Cu^{+2}} + E^\circ_{Ag^+/Ag}$

(2) $E^\circ_{cell} = E^\circ_{Ag^+/Ag} - E^\circ_{Cu^{+2}/Cu}$

(3) $E^\circ_{cell} = E^\circ_{Cu/Cu^{+2}} - E^\circ_{Ag/Ag^+}$

(4) $E^\circ_{cell} = E^\circ_{Ag/Ag^+} - E^\circ_{Cu^{+2}/Cu}$

EC0198

20. Find the incorrect formula for Daniell cell

(1) $E_{cell} = E^\circ_{cell} - \frac{2.303\,RT}{nF}\log_{10} Q$

(2) $\Delta G = \Delta G^\circ + RT \ln Q$

(3) $K_{eq} = 10^{\frac{nE^\circ_{cell}}{0.0591}}$

(4) At equilibrium $(\Delta G^\circ)_{T,P} = 0$

EC0199

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21. $\Lambda_m^\circ NaCl = z\,S\,cm^2\,mol^{-1}$

$\Lambda_m^\circ CH_3COONa = y\,S\,cm^2\,mol^{-1}$

$\Lambda_m^\circ HCl = x\,S\,cm^2\,mol^{-1}$

$\Lambda_m^c CH_3COOH = w\,S\,cm^2\,mol^{-1}$

When its concentration is 0.1 M then pick the correct option :

(1) $\Lambda_m^\circ CH_3COOH = x + y + z$

(2) $\alpha_{CH_3COOH} = \dfrac{w}{x+y-z}$

(3) $K_a = 0.1\left[\dfrac{w}{x+y-z}\right]$ if $\alpha << 1$

(4) $\Lambda_m^\circ CH_3COOH = \dfrac{\Lambda_{eq}^\circ CH_3COOH}{2}$

EC0200

22. The emf of the cell : Pt(s) | Br^- (0.01M) | Br_2(l) || H^+(0.001 M) | H_2(g) | Pt(s) will be

[Given : $E^\circ_{Br_2/Br^-}$ = 1.09 V $E^\circ_{H^+/H_2}$ and = 0V]

(1) –0.79 V (2) –1.39 V

(3) –1.30 V (4) –1.90V

EC0201

23. Given $E^0_{K^+/K}$ = –2.93 V, $E^0_{Ag^+/Ag}$ = 0.80 V, $E^0_{Hg^{+2}/Hg}$ = 0.79 V, $E^0_{Mg^{+2}/Mg}$ = –2.37 V, $E^0_{Cr^{+3}/Cr}$ = –0.74 V. Arrange these metals in increasing order of reducing power

(1) K, Mg, Cr, Hg, Ag (2) Ag, Hg, Cr, Mg, K

(3) Mg, K, Cr, Ag, Hg (4) Ag, Hg, K, Cr, Mg

EC0202

24. 4.5 gm of aluminium metal is deposited at cathode from Al^{+3} ions by a certain quantity of electric charge. Find volume of $H_{2(g)}$, produced at S.T.P. from H^+ ions by the same quantity of electric charge?

(1) 5.6 litre (2) 11.2 litre

(3) 22.4 litre (4) 44.8 litre

EC0203

25. The molar conductivity of 0.02 M HCOOH is 40.4 S cm^2 mol^{-1}, If $\Lambda_m^\circ(H^+)$ = 350 S cm^2 mol^{-1} and $\Lambda_m^\circ(HCOO^-)$ = 54 S cm^2 mol^{-1} then dissociation constant of HCOOH is –

(1) 2.22×10^{-3} (2) 2.22×10^{-4}

(3) 2.22×10^{-5} (4) 1.22×10^{-2}

EC0204

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EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	2	1	2	1	3	3	1	1	1	1	1	3	4	4
Que.	16	17	18	19	20	21	22	23	24	25					
Ans.	3	1	3	4	4	2	2	2	1	2					